Structures and Semantics of Equatives in Korean*

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Yeom, Jae-Il. 2018. **Structures and Semantics of Equatives in Korean.** *Korean Journal of Linguistics*, 43-2, 249-274. In this paper, I discuss equative constructions with *mankhum*. They are different from comparatives in three ways. First, in a clausal comparison construction, *pota* is always preceded by a *kes*-phrase, while *mankhum* is preceded by an adnominal clause. Second, *pota* can be used with a differential degree, while *mankhum* cannot. Third, *mankhum* can be used with a clause without a gradable verb, still denoting a degree, but *pota* cannot. These properties are related to the observation that in Korean, when an adnominal clause denotes a property of degrees, a Deg(ree)P(hrase) shows the property of a syntactic movement of an adjunct to the Spec of a CP. This motivates the idea that a degree is related to a situation. I also deal with quantificational DegP. An adnominal clause that denotes a degree with no gradable verb is a case of a coerced relative clause. A coerced relative clause is normally analyzed by inserting a contextually given relation between a situation described by the clause and the denotation of the head noun. This further motivates the adjunct analysis of a DegP. **(Hongik University)**

Key words: *mankhum*, equative, coerced relative clause, degree adjunct, quantification over degrees

1. Introduction

There is not much previous research on equatives in Korean.[1] Equatives are characterized by the use of *mankhum*. The morpheme is also used in other constructions than equatives. This provides some clues about how to analyze equatives and I will propose an analysis of equatives in which a *mankhum*-phrase is assumed to be an adjunct. To do this, I need to compare equatives with comparatives in Korean. I will start with

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1 Oh (2010) discusses all kinds of comparatives and equatives in Korean, but they are mostly descriptively. And she does not discuss cases where *mankhum* is used with an adnominal clause. I do not discuss all features of equatives with *mankhum*, and try to give a formal analysis of them, only focused on part of them.

three comparatives and equatives in English:

(1) a. John is taller than 6 ft.
b. John is taller than Mary.
c. John is taller than Mary is.

(2) a. John is as tall as 6 ft.
b. John is as tall as Mary.
c. John is as tall as Mary is.

In comparatives and equatives, the standard markers *than* and *as* take various phrases. (1a) is a case where *than* is followed by an expression like *6 feet*, which is called a measure phrase. In (1b), *than* is followed by an individual-denoting NP. In (1c), *than* is followed by a finite clause. For each type of comparative, there is a corresponding equative construction with the same structure. I will call comparative and equative constructions together comparison constructions. In the comparative constructions, the gradable adjective *tall* is neutral in the sense that it is not assumed that John and Mary are not tall.

(1b) and (2b) are phrasal comparison constructions and (1c) and (2c) are clausal comparison constructions. Generally, a phrasal comparative is characterized as a construction in which the standard marker like *than* or *as* is followed by a non-CP phrase (often, a NP or PP) that denotes a non-degree semantic entity. In a clausal comparative, the standard marker is followed by a CP that denotes a degree. See Hankamer (1973), Heim (1985), Kennedy (1997), etc.

The comparison constructions in Korean do not show such parallelisms. Here are the Korean comparison constructions:[2]

(3) a. ??Mino-nun 6 ft-pota (te) khu-ta.
-top -than more tall_V-dec[3]
'Mino is taller than 6 feet.'
b. Mino-nun Yuna-pota (te) khu-ta.

2 In this paper I use the following abbreviations: acc(usative case), adn(ominal ending), adn(ominal ending)m(odal), c(o)mp(lementizer), dec(larative), gen(itive case), hor(tative), imp(er)f(ective aspect), int(errogative), mod(ality), n(or)m(ina)l(izer), nom(inative case), p(a)st, p(e)rf(ect), p(er)f(ect)st(ate), pl(ural), top(ic marker).

3 In Korean, there is no adjective that can be used as the main predicate of a clause. If a gradable verb in Korean corresponds to the combination of the copula and an adjective in English, I will use the adjective with "$_V$" in the gloss.

−top −than more tall$_V$−dec
'Mino is taller than Yuna.'
c. (?)Mino−nun Yuna−ka khu−n kes−pota (te) khu−ta.
−top −nom tall$_V$−adn thing−than more tall$_V$−dec
'Mino is taller than Yuna is.'
(4) a. ??Mino−nun 6 ft−mankhum khu−ta.
−top −as tall$_V$−dec
'Mino is as tall as 6 feet.'
b. Mino−nun Yuna−mankhum khu−ta.
−top −as tall$_V$−dec
'Mino is as tall as Yuna.'
c. (?)Mino−nun Yuna−ka khu−n mankhum khu−ta.
−top −nom tall$_V$−adn as tall$_V$−dec
'Mino is as tall as Yuna is.'

In comparatives, the comparative morpheme *te* is optional in most cases. In (3b) and (4b), the standard markers, *pota* and *mankhum*, are preceded by the individual−denoting NP *Yuna*, and the comparison constructions are phrasal. In (3a) and (4a), however, the standard markers are not preceded by a measure phrase. Moreover, (3c) and (4c) do not simply compare Mino and Yuna's heights. They carry the meaning that Yuna and Mina are tall. This is not the way that phrasal comparison construction in Korean or a clausal comparison constructions in English are interpreted. These properties are observed in both comparison constructions, and they can be considered typological differences between Korean and English. I am more interested in non−typological differences.

A first non−typological difference is observed in the clausal comparison constructions in (3c) and (4c). In (3c) the standard marker is preceded by a *kes*−phrase, but in (4c), the standard marker is preceded by an adnominal clause. In both sentences, the degree of Yuna's height is determined by the adnominal clause *Yuna−ka khu−n*, and I take them to be clausal comparison constructions.[4] As a standard marker, *pota* is a more typical standard marker in that in all other constructions, it behaves like a P. If *mankhum* is preceded by an adnominal clause, it should be considered a nominal item. This is contrary to the expectation that a standard marker is generally a P. I will discuss why.

Second, *mankhum* can never take a measure phrase, while there are

[4] See Yeom (2016) for the analysis of Korean comparatives.

cases where *pota* can take a measure phrase:

(5) Mino−nun 6 ft−pota 1 in. (te) khu−ta.
−top −than more tall$_V$−dec
'Mino is 1 inch taller than 6 feet.'

Here, *te* 'more' comes with a measure phrase as a differential degree, in which case the comparative is acceptable. An equative construction does not have such a use because it does not require a differential degree due to its semantics. This is related to the fact that in an equative, no expression is used that corresponds to *te* 'more' in a comparative.

Third, *mankhum* can be used with a clause in non−comparison constructions. The corresponding comparatives are not acceptable:

(6) Yuna−nun motwu−ka chac−ul mankhum inki.iss−ta.
−top everyone−nom look.for−adn as popular$_V$−dec
'Yuna is popular to the extent that everyone wants her.'
(7) Yuna−nun haksayng−i−n mankhum kongpwu−ey
−top student−be−adn as study−at
cennyemhay−ya.ha−n−ta.
concentrate−must−dec
'Since Yuna is a student, she must concentrate herself on studying.'

The two uses in (6) and (7) are different from typical equatives in that the adnominal clauses that modify *mankhum* do not include a gradable verb. In (6), however, still the *mankhum*−phrase denotes a degree. In this respect, we need to explain how the *mankhum*−phrase denotes a degree. In (7), even the main clause does not include a gradable verb and the *mankhum*−phrase does not denote a degree. It is just an adjunct that expresses a reason for the event described in the main clause.

If *mankhum* is replaced with *kes−pota te*, the comparatives are odd:

(8) *Yuna−nun motwu−ka chac−ul kes−pota te inki.iss−ta.
−top everyone−nom look.for−adn thing−than more popular$_V$−dec
'Yuna is more popular than everyone wants her.'
(9) *Yuna−nun haksayng−i−n kes−pota te kongpwu−ey
−top student−be−adn thing−than more study−at
cennyemhay−ya.ha−n−ta.
concentrate−must−dec

'More than Yuna is a student, she must concentrate herself on studying.' (Lit.)

This indicates that *pota* is used only in comparatives.

I pointed out three differences between comparatives and equatives.[5] In this paper, I claim that a Deg(ree)P(hrase) is an adjunct and a degree is related to the event denoted by a gradable verb. This is a new observation which leads to a new analysis of comparison constructions. This claim is furthermore supported by the fact that *mankhum* can be used as a N and a P, and the fact that a *mankhum*-phrase can be used in non-equatives. The former fact is related to the observation that *mankhum* is preceded by an adnominal clause.

This paper is organized as follows. In Section 2, I claim that a DegP undergoes a syntactic movement to the Spec of the CP, which is a property of adjuncts, and that a *mankhum*-phrase is a PP. In Section 3, I show how a *mankhum*-phrase is interpreted as a degree adjunct when *mankhum* is modified by an adnominal clause with a gradable verb and when a clause denotes a degree without a gradable verb. I consider the latter case as a type of coerced relative clause. An analysis of a DegP as an adjunct is problematic in dealing with a quantificational degree expression. I propose a solution. In Section 4, I briefly discuss a case where *mankhum* is used with a clause that does not denote a degree. In Section 5, I summarize the discussions in the paper.

2. Syntactic and Semantic Properties of *Mankhum*

2.1. *Mankhum*-phrases as Adjuncts

There are various analyses of comparison constructions in English, but it is generally assumed that a degree is directly related to a gradable verb, whether a DegP is syntactically an argument or not.[6] Heim (1985, 2000)

[5] One more difference is that sub-equatives are generally accepted, while subcomparatives are not. This issue requires a separate paper.

[6] In the standard theory of a comparison construction, a gradable predicate is a relation between an individual and a degree, as in Cresswell (1977), von Stechow (1984), Rullmann (1995), Heim (1985, 2000), and others. In this analysis, a degree-denoting phrase is an argument of a gradable predicate. On the other hand, Bartch and Vennemann (1972) and Kennedy (1997) propose that a gradable predicate is a measure function that takes an individual and yields a degree. In this analysis, a degree-denoting phrase is an argument

assumes that a DegP is an argument of a gradable predicate, and that it is quantifier-raised. In this type of analysis, the LFs of (2b) and (2c) and their interpretations are as follows:[7]

(10) a. LF of (2b): John is [[as as Mary]$_i$ [t_i tall]]

b. $\llbracket as_1 \rrbracket = \lambda y\lambda G\lambda x[max(\lambda d[G(x,d)] \geq max(\lambda d'[G(y,d')]))$

(G: gradable verb; x, y: individuals;

$max(P) := \iota d[P(d)\ \&\ \forall d'[P(d') \rightarrow d' \leq d]])$

c. $\llbracket (2b) \rrbracket^{wg} = \llbracket as \rrbracket(\llbracket Mary \rrbracket)(\llbracket tall \rrbracket^{wg})(\llbracket John \rrbracket)$

$= max(\lambda d[tall'^{w}(j,d)]) \geq max(\lambda d'[tall'^{w}(m,d')])$

(11) a. LF of (2c): [as as [Op$_j$ [Mary is t_j tall]]]$_i$ [John is t_i tall]]

b. $\llbracket as_2 \rrbracket = \lambda d\lambda P[max(P) \geq d]$

c. $\llbracket [Op_i\ [Mary\ is\ t_i\ tall]] \rrbracket^{wg} = max(\lambda d[tall'^{w}(m,d)])$

d. $\llbracket (2c) \rrbracket^{wg}$

$= \llbracket as \rrbracket(\llbracket [Op_i\ [Mary\ is\ t_i\ tall]] \rrbracket^{wg})(\lambda d \llbracket John\ is\ t_i\ tall \rrbracket^{wg[d/i]})$

$= [max(\lambda d[tall'^{w}(j,d)]) \geq max(\lambda d'[tall'^{w}(m,d')])]$

For the two constructions, we need to assume that *as* is ambiguous. In (2b), *as* takes an individual-denoting NP as the standard of comparison, then a gradable adjective, and then another individual-denoting NP as the target of comparison, and the meaning is defined in (10b). The gradable verb applies to two individuals to yield the two maximal degrees of them with respect to the property denoted by the gradable verb. The compositional interpretation of (2b) is given in (10c). In (2c), *as* takes a degree-denoting clause and an individual-denoting NP. The meaning is defined in (11b). The DegP headed by *as* is quantifier-raised, and the degree-denoting clause in it is interpreted with the meaning of maximality, as in (11c), following von Stechow (1984) and Rullmann (1995). Thus the complement clause of *as* in (11) denotes the maximal degree of Mary's height. The quantifier-raised phrase takes the rest of the sentence, which denotes a property of degrees. Heim (2000) claims that

of an expression other than a gradable predicate. In either way, a degree is directly related to a gradable predicate.

[7] The concrete interpretations depend on the syntactic structures. I am assuming that in English a comparison morpheme combines with the standard of comparison first. But this is not required in the paper. We can assume a structure in which a comparison morpheme combines with a gradable predicate first. And considering the three constructions in (2), we need to define three meanings of *as*. In the paper, I only give the meaning relevant to the discussion at each part of the paper.

the main clause is also interpreted with the meaning of maximality. The meaning of (2c) is given in (11d). In the semantics of English equatives, it is assumed that a DegP is an argument of a gradable predicate.

For a maximal degree to be defined, we must assume the denotation of a gradable predicate is monotone in the sense of Heim (2000):

(12) A function f of type <d,et> is *monotone* iff
$$\forall x \forall d \forall d'[f(d)(x) \ \& \ d' < d \rightarrow f(d')(x)]$$

That is, if John is 180 cm tall, it is also true that he is 170 cm tall. Based on this assumption, a DegP can quantify over degrees.

However, there is syntactic evidence that a DegP is an adjunct in Korean. To show this, I will consider three types of adnominal clauses, following Yeom's (2015) discussions of adnominal clauses. One is an adnominal clause that abstracts over arguments. In this case, there is no island effect, which is the main characteristic of this type of adnominal clause. In (13a), the gap of a NP argument is embedded in another relative clause. A PP argument patterns with a NP argument with respect to island constraints, as in (13b). The verb *cwu* 'give' requires a PP argument with a recipient role, but the relative clause associated with the gap of that argument is embedded in another relative clause:

(13) a. [[e_i e_j chacaka−n_j kos−i] kacang me−n_i salam−un] Yuna−i−ta.
visit−adn place−nom most far$_V$−adn person−top −be−dec
'The person who$_i$ [the place she$_i$ visited] is farthest is Yuna.'
b. Mino−ka cwu−n chayk−i kacang ki−n salam−un
−nom give−adn book−nom most long$_V$−adn person−top
Yuna−i−ta.
−be−dec
'The person who$_j$ [the book Mino gave e_i to her$_j$] is longest was Yuna.'

Another type of adnominal clause is one denoting a property of times, places or instruments, which are generally considered adjuncts:

(14) Mino−nun Yuna−ka cha−lul kkulhi−ess−ta−ko sayngkaktoy−n
−top −nom tea−acc boil−pst−dec−cmp be.thought−adn
{ttay−ey, kos−eyse, yongki−ey} son−ul tey−ess−ta.
{time−at, place−at, kettle−to} hand−acc be.burnt−pst−dec
'Mino was burnt {when, where, to the kettle} it was thought that Yuna made tea.'

(15) *Mino−nun cha−lul kkulhi−n salam−i celm−un {ttay−ey,
−top tea−acc boil−adn person−nom young$_V$−adn {time−at,
kos−eyse, yongki−ey} son−ul tey−ess−ta.
place−at, kettle−to} hand−acc burnt$_V$−pst−dec
'Mino got burnt {when, where, to the kettle} the person who made tea was young.'

The adnominal clauses cannot denote a property of times, places or kettles for boiling tea. The relation between the gaps and the adnominal clauses are not clause−bound, but subject to island constraints.

The last type of adnominal clauses is one denoting a property of reasons or purposes, which are also considered adjuncts:

(16) Mino−nun Yuna−ka nuc−ess−ta−ko sayngkaktoy−nun
−top −nom late$_V$−pst−dec−cmp be.thought−adn
{iywu, *mokcek}−(l)ul ic−ess−ta.
{reason, purpose}−acc forget−pst−dec
'Mino forgot the {reason, *purpose} that Yuna was thought to be late.'

Here the reason is not for Yuna being late, but for Yuna being thought to be late. We cannot think of a purpose for being thought. Instead, the purpose should be related to Yuna being late, but it is not because Yuna's state is described by the embedded clause. This shows that if an adnominal clause denotes a property of reasons/purposes, it should be related to the situation described by the topmost clause of the clause.

We saw three types of adnominal clauses, depending on the syntactic restrictions between gaps relevant and the adnominal clauses. Then what type of adnominal clause does a degree−denoting clause belong to? It patterns with time, place and instrument adnominal clauses. A degree can be related to a predicate that is embedded in the complement clause of an attitude verb, as in (17), but a degree−denoting clause is subject to island constraints, as in (18) and (19).[8] In (18), the degrees of the visitor's happiness and Mino's happiness are compared. It does not mean that Mino was happy because a happy person visited him.

8 Pinkham (1982) and Kennedy (1997, 2002) showed that English comparatives are also subject to island constraints.

(17) Mino−nun Yuna−ka hayngpokha−ta−ko sayngkaktoy−nun
−top −nom happy$_V$−dec−cmp be.thought−adn
mankhum hayngpokha−ta.
as happy$_V$−dec
'Mino is as happy as Yuna is thought to be.'
(18) *Mino−nun hayngpokha−n salam−i chacao−n mankhum
−top happy$_V$−adn person−nom visit−adn as
hayngpokha−ta.
happy$_V$−dec
'Mino is as happy as a person who is happy visited him.'
(19) *Mino−nun Yuna−ka hayngpokha−l ttay ttena−n mankhum
−top −nom happy$_V$−adn time leave−adn as
hayngpokha−ta.
happy$_V$−dec
'Mino is as happy as Yuna left when she is.'

A time, place or instrument for an event is normally expressed with a PP. Even if realized as an NP, the NP is analyzed as a PP, following Larson (1985) and Emonds (1987). So a DegP must be taken as an adjunct PP that moves to the Spec of a CP at LF, subject to island constraints.[9]

(20) a. A DegP used in relation to a gradable verb is an adjunct PP.
b. A DegP moves to the Spec of a CP at LF.

2.2. *Mankhum*−phrases are PPs

We saw that a DegP in a degree−denoting clause is an adjunct. Then a *mankhum*−phase in an equative should be an adjunct. This structural correspondence is shown here:

(21) a. $\cdots[\cdots[_{CP} \cdots[_{VP}\ [t_i]_{DegP}\ V_{gradable}]$−un $Op_i]_{CP}$ kes−pota $te]_{DegP}\ V_{gradable}$
b. $\cdots[\cdots[_{CP} \cdots[_{VP}\ [t_i]_{DegP}\ V_{gradable}]$−un $Op_i]_{CP}$ $mankhum]_{DegP}\ V_{gradable}$

In this structure, $[t_i]_{DegP}$ is the trace of a DegP left by the movement to the position of Op_i. If the DegPs used with a gradable verb in the adnominal clauses are adjuncts, the *mankhum*−phrase and *te*−phrase are

[9] For an appropriate interpretation, the P of an adjunct PP must be restructured, following von Stechow (1996), as in all other pied piped PPs.

also adjuncts in the matrix clauses. Note that *mankhum* is preceded by an adnominal clause. It is a property of a N. However, if *mankhum* were a N, a *mankhum*−phrase would be a NP. In this subsection, I will show that a DegP is rather a PP.

To begin with, *mankhum* as a P is well established, given that it is used as a standard marker in phrasal equatives, as in (4b). The following examples also support this claim:

(22) Yuna−nun pakk−eyse−mankhum an−eyse coyongha−ta.
−top outside−at−as inside−at quiet$_V$−dec
'Yuna is as quiet indoors as outdoors.'
(23) a. Yuna−nun kay−lul ilh−un kes−mankhum i il−i
−top dog−acc lose−adn thing−as this thing−nom
sulph−ess−ta.
sad$_V$−pst−dec
'Yuna was as sad about this thing as the event that she lost her dog.'
b. i cha−nun ney−ka sa−n kes−mankhum ppalu−ta.
this car−top you−nom buy−adn thing−as fast$_V$−dec
'This car is as fast as what you bought.'

If the phrases before *mankhum* denote other things than degrees, *mankhum* is definitely a P, whether it is preceded by a clause or not.

Even in other clauses than clausal equatives, *mankhum* can be considered a P: in (7), a *mankhum*−phrase is an adjunct. In (4c) and (6), where the *mankhum*−phrases denote degrees, *mankhum* can be replaced with *cengto−lo* 'degree−to', which forms a PP:

(4.c') (?)Mino−nun Yuna−ka khu−n cengto−lo khu−ta.
−top −nom tall$_V$−adn degree−to tall$_V$−dec
'Mino is as tall as Yuna is.'
(6') Yuna−nun motwu−ka chac−ul cengto−lo inki.iss−ta.
−top everyone−nom look.for−adn degree−to popular$_V$−dec
'Yuna is popular to the extent that everyone wants her.'

And (4.c') is not quite natural, but it is acceptable if (4c) is acceptable, and (6') is perfect. This shows that whenever a *mankhum*−phrase denotes a degree, *mankhum* can be replaced with *cengto−lo* 'degree−to'. This indicates that a *mankhum*−phrase in an equative occurs in a PP position.

On the other hand, *mankhum* can also be used as a N:

(24) A: ton-i elmana philyoha-nya?
money-nom how.much necessary-int
'How much money do you need?'
B: [$_{NP}$ cip-ul sa-l mankhum]-i philyoha-ta.
house-acc buy-adn as-nom necessary-dec
'I need the amount of money to buy a house.'

The *mankhum*-phrase is used as a NP, and *mankhum* is the head. *Mankhum* can also head a degree-denoting NP_{Deg}, without a P. For example, a *mankhum*-phrase can be used as a differential DegP, as in (25). Note that a measure phrase, which is considered an NP, is used in the position of a differential DegP. In (25), *mankhum* cannot be replaced with *cengto-lo* 'degree-to'.

(25) thap-un kenmwul-i kiwul-e.iss-nun mankhum te kiwul-e.iss-ta.
tower-top building-nom $tilt_V$-impf-adn as more $tilt_V$-impf-dec
'The tower is more slanted by the degree that the building is.'

In Korean, there are NPs used in an adjunct position. Expressions like *ttay* 'time', *tey* 'place', *yang* 'as if', etc. are nouns. They can be modified by an adnominal clause, as in (26)-(28). However, the NPs are used as an adjunct, just like *mankhum*-phrases in equatives.

(26) Mino-ka o-l ttay na-nun ttena-n-ta.
-nom come-adn time I-top leave-impf-dec
'When Mino comes, I will leave.'
(27) Mino-lul seltukha-nun tey ithul-i keli-ess-ta.
-acc persuade-adn place two.days-nom take-pst-dec
'It took two days to persuaded Mino.'
(28) Yuna-nun el-i ppaci-n yang se.iss-ess-ta.
-top mind-nom leak-adn as.if stand-pst-dec
'Yuna was standing as if she lost her mind.'

There are two differences between *mankhum* and expressions like *ttay* 'time', *tey* 'place', *yang* 'as if', etc. First, the latter expressions are never used as a P preceded by a NP, while *mankhum* can be, as in (22) and (23). Second, the latter can be used with an overt P: ⋯ *ttay-ey* '⋯ time-at ', ⋯ *tey-ey* ' ⋯ place-at', ⋯ *yang-ulo* '⋯ way-in', etc. But a

mankhum-phrase as an adjunct is never used with a separate P:

(29) Inho-nun cwuk-ul {cengto-lo, mankhum-(*ulo)} phikonhay-ss-ta.
-top die-adnm {degree-to, as-to } $tired_V$-pst-dec
'Inho was tired to the degree that he would die.'

If a noun *cengto* 'degree' is used instead of *mankhum*, the P *(u)lo* is necessary. These observations show that even when *mankhum* is modified by an adnominal clause, it maintains the status of a P so that no other P can be allowed. Therefore, it is a peculiar property that a *mankhum*-phrase behaves like a PP but *mankhum* is preceded by an adnominal clause. To capture the fact that a *mankhum*-phrase behaves like a NP phrase-internally but behaves like a PP phrase-externally and that it behaves in a different way than those in (26)-(28), I will assume the following structure:

(30) [$_{PP}$ [$_{NP}$ [$_{CP}$ ⋯-ul/un] [∅]$_N$]-mankhum]

We can suppose that the null noun is something like *cengto* 'degree' because *mankhum* is always preceded by the noun:

(31) a. Inho-nun cwuk-ul cengto-mankhum phikonhay-ss-ta.
-top die-adnm degree-as $tired_V$-pst-dec
'Inho was tired to the extent that he would die.'
b. Inho-nun Mina-ka ttokttokha-n cengto-mankhum ttokttokha-ta.
-top -nom $smart_V$-adn degree-as $smart_V$-dec
'Inho is as smart as Mina is.'

Cengto 'degree' can be inserted even when *mankhum* is used as a N:

(24') [$_{NP}$ cip-ul sa-l cengto mankhum]-i philyoha -ta.
house-acc buy-adn degree as-nom necessary-dec
'The amount (of money) to buy a house is necessary.'

Thus when *mankhum* is used as a P, we can still assume that a noun like *cengto* 'degree' is omitted.[10] It is omitted because it is redundant: it

[10] To account for the fact that *mankhum* is used as a P or N and that when it is used as a P, no other P is allowed, we assume the following syntactic structure:

has the same meaning as *mankhum*.

3. Interpretation of *Mankhum*−phrases with a Clause

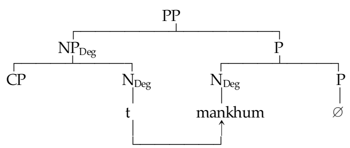
3.1. Degrees in Relation to Events

A phrase denoting a time, place or instrument is related to an event or situation. To capture the relation of an event/situation to a temporal expression, for example, I need to make some preliminary assumptions, which are considered standard ones. A VP denotes a property of events or states, which I will call simply events. An aspect changes it into a property of times, by specifying the inclusion relation (i.e., $\subset$) between the event time (i.e., $\tau(e)$) and the reference time. Other times than the utterance time are represented as t, t', t", ⋯. In order to be interpreted properly, a time−denoting phrase is adjoined to a VP or an AspP. If a time adverbial is adjoined to a VP, it restricts an event time, and if a time−adverbial is adjoined to an AspP, it restricts a reference time, which is determined by the tense of a sentence. A tense specifies the precedence relation between the reference time and the utterance time u, but I will ignore Tense:

(32) $\llbracket [_{VP}\ \alpha] \rrbracket^{wg} = \lambda e[\llbracket \alpha \rrbracket(e)]$
$\llbracket [_{AspP}\ \alpha\ \beta_{Asp}] \rrbracket = \lambda w \lambda t \exists e[\llbracket \alpha \rrbracket^{w}(e)\ \&\ T(t, \tau(e))]$
($T \in \{\subset, \supset\}$; $\tau(e)$: the event time of e; t: reference time)

If the predicate is stative, $t \subset \tau(e)$; otherwise, $t \supset \tau(e)$. This is not

i.



This structure obviates the use of a head noun modified by an adnominal clause, separately from *mankhum*. Since *mankhum* occurs in the position of a P, no other P is necessary. A head movement is a peculiar property of a lexical item. This applies to *mankhum* but not to *cengto*, though both can be used as a N with the same meaning.

sufficient to capture all cases, but I will not go into this issue furthermore.

To interpret an equative, I need to give the semantics of the structure given in (30). The empty noun ∅ is semantically null except for the restriction that it is a property of degrees. The restriction is not part of the entailment because if it is not a property of degrees, the sentence does not become false but odd. And *mankhum* is interpreted as a standard marker. Since a *mankhum*−phrase forms an adjunct phrase, its meaning needs to be defined with respect to an event:

(33) a. $\llbracket \varnothing \rrbracket^{w} = \lambda P \lambda d[P(d)]$, where d ranges over degrees.
b. $\llbracket \text{mankhum} \rrbracket^{w} = \lambda P \lambda e[\delta(e) \geq \max(P)]$ $(= \llbracket \varnothing\text{−mankhum} \rrbracket^{w})$

An event can be connected to a degree by using the operator δ, which is a function that takes an event denoted by a gradable verb and yields the degree of the property denoted by the verb. Since ∅ is semantically null, "∅−*mankhum*" means the same as *mankhum*.

Since we are interested in the interaction between an event and an adjunct, I will discuss a time adverbial adjoined to a VP first:

(34) Mino−nun$_i$ [[[ecey [t_i ttena]$_{VP}$]$_{VP}$−$\varnothing_{Asp}$]$_{AspP}$]−ss]$_{TP}$−ta.
−top yesterday leave−asp−pst−dec
'Mino left yesterday.'

(35) $\llbracket [t_i\ \text{ttena}]_{VP} \rrbracket^{wg} = \lambda e[\text{leave}^{w}(e,g(i))]$
$\llbracket \text{ecey} \rrbracket^{wg,u}$
$= \lambda e \exists t'[\exists t''[\text{day}^{w}(t')\ \&\ \text{day}^{w}(t'')\ \&\ t' \propto t''\ \&\ t'' \subseteq u]\ \&\ \tau(e) \subseteq t']$
$= \lambda e \exists t'[\text{yesterday}(t')\ \&\ \tau(e) \subseteq t']$, where
(i) $\text{yesterday}(t') = \exists t''[\text{day}^{w}(t')\ \&\ \text{day}^{w}(t'')\ \&\ t' \propto t''\ \&\ t'' \subseteq u]$
(ii) $a \propto b$: 'a butt(s) b in that order'; u: utterance time.
$\llbracket [\text{ecey}\ [t_i\ \text{ttena}]] \rrbracket^{wg} = \lambda e \exists t'[\text{leave}^{w}(e, g(i))\ \&\ \text{yesterday}(t')\ \&\ \tau(e) \subseteq t']$

The operator τ connects the event denoted by the VP to the time denoted by the time adverbial. The meaning of *ecey* is given as $\lambda e \exists t'[\text{yesterday}(t')\ \&\ \tau(e) \subseteq t']$. The two properties of events are conjoined.

Similarly, (4c) can be assumed to have the LF in (36), which is interpreted as in (37).[11]

[11] It is possible that the DegP moves to the Spec of CP together with the null P, but for a proper interpretation, the null P must be reconstructed.

(36) Mino−nun$_i$ $[[[_{CP}$ $[$Yuna−ka$_j$ $[[_{PP}\ t_k]\ \varnothing_P]\ [\ t_j$ khu$]_{VP}]_{VP}]$−n $Op_k]_{CP}]$
−top −nom tall$_V$−adn
mankhum$]$ $[t_i$ khu$]_{VP}]_{VP}$−ta.
as tall$_V$−dec

(37) a. $[\![\,[t_j \text{ khu}]\,]\!]^{wg} = \lambda e[tall^w(e,g(j))]$
b. $[\![\,[_{PP}\ [\alpha]_{DegP}\ \varnothing_P]\,]\!]^{wg} = \lambda e[\delta(e) =_{id} [\![\,\alpha\,]\!]^{wg}]$
(δ: a function taking an event and yielding a degree)
c. $[\![\,[[_{PP}\ t_k\ \varnothing_P]\ t_j \text{ khu}]\,]\!]^{wg} = \lambda e[tall^w(e,g(j))\ \&\ \delta(e) =_{id} g(k)]$
d. $[\![\,[_{CP}\ [\text{ Yuna−ka } [[_{PP}\ [t_k]_{DegP}\ \varnothing_P]\ t_j \text{ khu}]_{VP}]_{VP}]\text{−n } Op_k]\,]\!]^{wg}$
$= \lambda d [\![\,[\text{Yuna−ka } [[_{PP}\ [t_k]_{DegP}\ \varnothing_P]\ t_j \text{ khu}]_{VP}]_{VP}]\,]\!]^{wg[d/k]}$
$= \lambda d \exists e[tall^w(e,y)\ \&\ \delta(e) =_{id} d]$

(37a) is the meaning of the smaller VP, which is a property of events. (37b) is the meaning of a degree adjunct, which consists of a DegP and a null P. δ connects the event to the degree. In (37c), the DegP is a trace and we get the meaning of the larger VP. Both the meanings of the smaller VP and the degree adjunct are properties of events, and they are conjoined. When the embedded AspP is interpreted, the event variable is bound by an existential quantifier. The meaning of the adnominal clause is an abstraction over degrees with respect to the property of being smart.

Next we need to interpret *mankhum*, the meaning of which is given in (32). The meaning of an adnominal clause corresponds to P in (32). Then the interpretation of (36) can continue as follows:

(37') $[\![\,[[_{CP}[\text{Yuna−ka } [[_{PP}\ t_k\ \varnothing_P]\ t_j \text{ khu}]_{VP}]_{VP}\text{−n } Op_k]]\ \varnothing\text{−mankhum}]\,]\!]^{wg}$
$= \lambda e[\delta(e) \geq max(\lambda d \exists e[tall^w(e,y)\ \&\ \delta(e) =_{id} d])$

Then the interpretation of the matrix clause continues as follows:

(37") a. $[\![\,[t_i \text{ khu}]\,]\!]^{wg} = \lambda e'[tall^w(e',g(i))]$
b. $[\![\,[[_{CP} \text{ Yuna−ka khu−n } Op_k] \text{ mankhum}]\ [t_i \text{ khu}]]\,]\!]^{wg}$
$= \lambda e'[tall^w(e',g(i))\ \&\ \delta(e') \geq max(\lambda d \exists e[tall^w(e,y)\ \&\ \delta(e) =_{id} d])]$
c. $[\![\,(36)\,]\!]^{wg}$
$= \exists e'[tall^w(e',m)]\ \&\ \delta(e') \geq max(\lambda d \exists e[tall^w(e,y)\ \&\ \delta(e) =_{id} d])]$

In (37'a), the meaning of the matrix VP is also a property of events, and the two are conjoined, as in (37"b). The event variable is existentially closed when the AspP is interpreted, as in (37"c). I will ignore the rest

of the interpretation.

In (37"d), the meaning of the sentence is given as the relation of "≥", but it is implicated that the two degrees are equal because the corresponding comparative is not used. So far I have shown how (4c) can be interpreted compositionally.

3.2. Coerced Degree−denoting Clauses

In this paper, I get a degree in relation to a situation or event. This not only follows from the fact that a DegP acts as an adjunct. It is required in explaining a sentence like (6), because only the main predicate is a gradable one. Since the adnominal clause that modifies *mankhum* does not include a gradable verb, the adnominal clause in the *mankhum*−phrase itself does not denote a degree. It only restricts the degree denoted by the matrix clause. Hence it needs to denote a property of degrees without any internal gradable verb.

In Korean, there is an adnominal clause denoting something that normally is not denoted by the clause. It is called a coerced relative clause in Yeom (2017).[12] One example is given here:

(38) sayngsen−i tha−nun naymsay
fish−nom burn−adn smell
'smell that a fish burns' (Lit.)

One of the properties that such a construction shows is that the thing denoted by the NP is related to the situation that the topmost clause describes, as pointed out in Yeom (2017). (39) is odd because the smell cannot come from the situation of thinking that a fish burns.

(39) ??sayngsen−i tha−n−ta−ko sayngkaktoy−nun naymsay
fish−nom burn−impf−dec−cmp be.thought−adn smell
'smell that it is thought that a fish burns' (Lit.)

We can make the same observation in clauses that denote a degree without a gap. In (40), the degree cannot be related to the situation of

[12] It is generally observed in East−Asian languages. See also Murasugi (1991), Matsumoto (1988, 1989, 1997) and Cha (1997), Tang (1979:243−289), Tsai (1997), Zhang (2008).

thinking that everyone wants her. This supports the idea that (6) can be treated as a coerced relative clause.

(40) ??Yuna-nun$_i$ motwu-ka pro$_i$ chac-nun-ta-ko
-top everyone-nom want-impf-dec-cmp
sayngkaktoy-l mankhum inki.iss-ta.
be.beieved-adn$_m$ as popular-dec
'Yuna is popular to the extent that everyone is believed to want her.'

Another property of the construction in (6) is that the adnominal ending is *-(u)l*, not *-(nu)n*. The ending generally conveys the meaning of modality, but this is not always the case, as pointed out in Ywu (2009):

(41) Mino-ka ttena-l ttay Yuna-ka o-ass-ta.
-nom leave-adnm time -nom come-pst-dec
'Yuna came when Mino left.'

In (41), there is no modality involved. In (6), the same ending used with *mankhum* does not carry the meaning of modality, either.

When a clause denotes a degree with no gradable verb, the situation the clause describes can be different from an event denoted by a VP:

(42) Mino-to col-ci anh-ul mankhum yenghwa-ka
-also doze-nml not.do-adnm as movie-nom
caymi.iss-ta.
interesting$_V$-dec
'The movie's interesting to the extent that even Mino didn't doze off.'

The VP *col* 'doze' is negated and there is no event of dozing. The degree of being interesting is related to the situation in which there is no event described by the embedded VP. This requires introducing a situation **that** is described by a negative clause, which is a larger phrase than a VP. For this purpose, we can distinguish an event e and a situation s, which is a part of a world (i.e., s⊂w): a VP denotes a property of events, and a clause denotes a property of situations. See Yeom (2015, 2017) for more detailed discussions. To introduce a situation, I need to assume that an aspect is interpreted as follows:

(43) $[\![[_{AspP}\ \alpha\ \beta_{Asp}]]\!]$

$= \lambda w \lambda s \lambda t \exists e[s \subset w\ \&\ [\![\alpha]\!]^{w}(e)\ \&\ \tau(e) \subseteq \tau(s)\ \&\ T(t, \tau(e))]$, where $T \in \{\subset, \supset\})$

We distinguish events and situations, but I suppose that they are of the same semantic type. Both an event and a situation are semantic objects that have a spatio-temporal location. In (43), the situation is the sum of the events described by the VP. Thus each event described by the VP is part of the situation described by the clause. If there is only one event, that event becomes the situation of the sentence. They are closed under the sum operation (i.e., +).

A coerced relative clause is interpreted as a process that the head noun coerces a certain relation between the denotation of the head noun and the situation described by the topmost clause. In (44), R is a relation between a situation s and a thing x. We are dealing with degrees, and x can be replaced with d.

(44) If $[\![[_{CP}\ \alpha]]\!]^{wg} = \lambda s[P(s)]$ and $[\![[_{N}\ \beta]]\!]^{wg} = \lambda x[Q(x)]$, then $[\![[_{NP}\ \alpha\ \beta]]\!]^{wg} = \lambda x \exists s[Q(x)\ \&\ P(s)\ \&\ R(x,s)]$

We can apply this to (38) as follows:

(45) $[\![\text{naymsay}]\!]^{w} = \lambda x[\text{smell}^{w}(x)]$
$[\![\text{sayngsen-i tha-nun}]\!]^{wg}$
$= \lambda s \exists y \exists e[s \subset w\ \&\ \text{fish}^{w}(y)\ \&\ \text{burn}^{w}(e,y)\ \&\ \tau(e) \subseteq \tau(s)]$[13]
$[\![\text{sayngsen-i tha-nun naymsay}]\!]^{wg}$
$= \lambda x \exists s \exists y \exists e[s \subset w\ \&\ \text{fish}^{w}(y)\ \&\ \text{burn}^{w}(e,y)\ \&\ \tau(e) \subseteq \tau(s)\ \&\ \text{smell}^{w}(x)\ \&\ R(x,s)]$
$= \lambda x \exists s \exists y[s \subset w\ \&\ \text{fish}^{w}(y)\ \&\ \text{burn}^{w}(s,y)\ \&\ \text{smell}^{w}(x)\ \&\ R(x,s)]$
(R: relation given by the context)

Since there is only one event, there is no distinction between s and e. There are restrictions on R, but I am concerned with them here.[14]

[13] In this meaning representation, we might need to include spatial inclusion relations, but for simplicity's sake I will ignore them.

[14] In some cases, there is a cause-effect relation between a situation described by the clause and a thing denoted by the head noun, as in (38). But there are other cases where a predicate introduced from the TELIC qualia of the head noun, in terms of Generative Lexicon by Pustejovsky (1995), is considered to connect the situation and the thing:

We can also think of a relation R that connects a situation and the degree with respect to the gradable predicate in the main clause. To do this, I interpret the main clause first:

(46) a. $[\![\text{inki.iss}]\!]^{w} = \lambda x\lambda e[\text{popular}^{w}(e,x)]$
 b. $[\![[_{VP}\ t_i\ \text{inki.iss}]]\!]^{wg} = \lambda e[\text{popular}^{w}(e,g(i))]$

Now we need to interpret the *mankhum*-phrase. A *mankhum*-phrase needs an adnominal clause, which is normally interpreted as a property of situations because there is no gradable verb in it. When the clause combines with ∅-*mankhum*, a relation R converts a property of situations to a property of degrees:

(47) $[\![\varnothing\text{-mankhum}]\!]^{w}$
 $= \lambda P\lambda e[\delta(e) \geq \max(P)]$ (P: a property of degrees)
 $[\![[\text{motwu-ka pro}_i\ \text{chac-ul}]]\!]^{wg}$
 $= \lambda s[s\subset w\ \&\ \forall x[\text{person}^{w}(x) \rightarrow \exists e[\text{want}^{w}(e,x,g(i))\ \&\ \tau(e)\subseteq\tau(s)]]]$
 $[\![[\text{motwu-ka pro}_i\ \text{chac-ul}\ \varnothing\text{-mankhum}]\!]^{wg}$
 $= [\![\text{mankhum}]\!]^{w}(\lambda d\exists s[s\subset w\ \&\ \forall x[\text{person}^{w}(x) \rightarrow$
 $\exists e[\text{want}^{w}(e,x,g(i))\ \&\ \tau(e)\subseteq\tau(s)]]\ \&\ R(d,s)])$
 $= \lambda e'[\delta(e') \geq \max(\lambda d\exists s[s\subset w\ \&\ \forall x[\text{person}^{w}(x) \rightarrow$
 $\exists e[\text{want}^{w}(e,x,g(i))\ \&\ \tau(e)\subseteq\tau(s)]]\ \&\ R(d,s)])$

One problem with this result is that the adnominal clause does not include a gradable verb, and we cannot determine what kind of degree should be derived. More seriously, when the *mankhum*-phrase is interpreted, the matrix gradable verb is not available. The only way to get the matrix gradable verb when *mankhum* is interpreted is to get it contextually. The main clause provides a salient gradable predicate G and its arguments e and y:

(48) $R(d,s) = [G(s+e,y)\ \&\ \delta(s+e)=d]$ (G, e, y given in context)

In the example at hand, the context provides a gradable predicate *inki.iss* 'be popular', the situation s and the individual argument g(i). The

i. meli-ka cohaci-nun chayk
 brain-nom get.better-adn book
 'a book that (if anyone reads it) they can get smarter'

NP roughly denotes a property of degrees of g(i) being popular when everyone wants g(i):

(49) $\lambda e'[\delta(e')\geq\max(\lambda d\exists s[s\subset w\ \&\ \forall x[\text{person}^{w}(x)\rightarrow\exists e[\text{want}^{w}(e,x,g(i))$
 $\&\ \tau(e)\subseteq\tau(s)]]\ \&\ \text{popular}^{w}(s+e,g(i))\ \&\ \delta(s+e) =_{id} d])]$

This combines with the meaning of the main clause in (46b) and we get (50a), which combines with the meaning of the subject to yield (50b):

(50) a. $[\![[\text{motwu-ka} \ldots \text{mankhum}\ t_i\ \text{inki.iss}]\!]^{wg}$
 $= \lambda e'[(i)\ \text{popular}^{w}(e,g(i))\ \&\ (ii)\ \delta(e')\geq\max(\lambda d\exists s[s\subset w\ \&$
 $\forall x[\text{person}^{w}(x) \rightarrow \exists e[\text{want}^{w}(e,x,g(i))\ \&\ \tau(e)\subseteq\tau(s)]]$
 $\&\ \text{popular}^{w}(s+e,g(i))\ \&\ \delta(s+e) =_{id} d])]$
 b. $[\![\text{Yuna-nun} \ldots \text{inki.iss}]\!]^{wg}$
 $= \lambda e'[(i)\ \text{popular}^{w}(e,y)\ \&\ (ii)\ \delta(e')\geq\max(\lambda d\exists s[s\subset w\ \&$
 $\forall x[\text{person}^{w}(x) \rightarrow \exists e[\text{want}^{w}(e,x,y)\ \&\ \tau(e)\subseteq\tau(s)]]$
 $\&\ \text{popular}^{w}(s+e,y)\ \&\ \delta(s+e) =_{id} d])]$

The rough meaning is that the degree of Yuna's popularity is the same as Yuna's degree of popularity in a situation where she is popular and everyone wants her. This is the reading we want.

3.3. Quantification over Degrees

One problem with analyzing a DegP as an adjunct is that when an adjunct is a negative word, the meaning we get is wrong:

(51) a. The pictures are to no degree different.
 b. $\exists e[\text{different}'^{w}(e,\iota x[\text{pictures}'^{w}(x)])\ \&\ \delta(e) =id\ 0]$

This interpretation would wrongly entail that the pictures are different.

To correct this interpretation, we need to interpret an adjunct as a quantifier. To get the idea, consider the following example:

(52) a. John found gold nowhere.
 b. $\exists e[\text{found}'^{w}(e,j,\text{gold}')\ \&\ \sigma(e) = \text{nowhere}'^{w}]$
 c. $[\![\text{nowhere}]\!]^{wg}(\lambda x[\![\text{John found gold}\ t_i]\!]^{wg[x/i]})$
 $= \neg\exists x\exists e[\text{place}'^{w}(x)\ \&\ \text{found}'^{w}(e,j,\text{gold}'^{w})\ \&\ \sigma(e) =_{id} x]$

I assume the operator σ that takes an event or situation and yields the place of that event/situation. If *nowhere* were not a quantifier and (52a) were interpreted as in (52b), it would be entailed that John found gold. Instead *nowhere* needs to be interpreted as a quantifier, as in (52c).

However, it is difficult to interpret a DegP as a quantifier because a degree is hard to quantify. If John is 175cm tall, 175cm is not considered a quantity. It is just one point of the scale of height. On the other hand, a degree is treated as a quantity:

(53) inho−nun **manhi** phikonhay−ta.
Inho−top **much** tired_V−dec
'Inho is very tired.'

In this example, *manhi* 'much' is used as meaning 'very'. For such cases, we need to find a way of capturing a degree as a quantity.

One way to quantify a degree is to use the assumption that the denotation of a gradable predicate is monotone, as in (12). That is, Inho is 175cm tall, then it is also true that he is 170 cm tall. Then (53) can be interpreted by really quantifying over degrees. The meanings of *manhi* 'much' can be defined as follows:

(54) a. $[\![\text{manhi}]\!]^w = \lambda P[\text{MUCH}(D)(\{d|\ P(d)\})]$, where
D is a whole set of degrees; P is a property of degrees.
b. $\text{MUCH}(D)(P) = 1$ iff $(\max(D)-\min(D)) \times (\#(P)/\#(D)) > n$.

D is a whole set of degrees in the scale of a property, and P is a property of degrees. Both are taken to be sets of degrees. Then MUCH(D)(P) is true iff (max(D)−min(D)) multiplied with the ratio of #(P)/#(D) is larger than a certain number.

This can apply to (53). If the scale of being tiredness is from 0 to 9, D = {0, 1, ⋯, 9}. If more than 6 is "much" in the scale of tiredness and Inho is 7 tired, then P determines the set {0, 1, ⋯, 7}. Then the truth conditions are $(9 - 0) \times 8/10 = 7.2 > 6$. The conditions are met and the sentence is true:

(55) a. Suppose that D = {0, 1, ⋯, 9}; Inho is 7 tired.
b. $\text{MUCH}(D)(P) = 1$ iff $(\max(D)-\min(D))\times(\#(P)/\#(D)) > 6$.

c. $[\![(C)]\!]^w$
$= \lambda P[\text{MUCH}(D)(\{d|P(d)\})](\lambda d[\text{tired}^w(e,i)\ \&\ \delta(e) = d])$
$= \text{MUCH}(D)(\{d|\ \text{tired}^w(e,i)\ \&\ \delta(e) = d\})]$, where
$\{d|\ \text{tired}^w(e,i)\ \&\ \delta(e) = d\} = \{0, 1, \cdots, 7\}$

The number we get as the truth−conditions is not exact because the range of the scale is discrete, but the basic idea is clear. In this way, a degree can be dealt with as a quantity.

4. Non−degree Use of *Mankhum*

I will briefly discuss a construction in which *mankhum* does not denote a degree, like the one in (7). This construction is not a comparison construction, despite the use of *mankhum*. To support this claim, I will briefly list some properties of the construction, in comparison with *mankhum*−constructions in which *mankhum* denotes a degree.

The example in (7) is repeated here. It differs from (6) in several ways.

(56) Yuna−nun haksayng−i−n mankhum kongpwu−ey
−top student−be−adn as study−at
cennyemhay−ya.ha−n−ta.
concentrate−must−dec
'Because Yuna is a student, she must concentrate herself on studying.'

First, the adnominal ending is −*(nu)n*, not −*(u)l*, and *mankhum* has a similar meaning to 'because' and does not have a comparison meaning. Second, the adnominal clause is not related to any degree. The clause denotes a situation, but it is not related to any degree in comparison with other alternative situations. And the main predicate is not gradable. Third, this construction shows some restrictions on possible moods. *Mankhum* can be used in a declarative, imperative, and hortative, but not in an interrogative. (57) is a question corresponding to (56):[15]

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[15] A reviewer points out that some interrogatives are allowed:

i. Yuna−nun haksayng−i−n mankhum kongpwu−ey cennyemha−ko.iss−keyss−ci?
−prog−mod−dec
'Yuna is concentrating on her study because she is a student, isn't she?'

(57) ??Yuna-nun haksayng-i-n mankhum kongpwu-ey
cennyemhay-ya.ha-nya?
-int

With a second person subject, only the imperative is used:

(58) ney-ka hyeng-i-n mankhum cham-{ala!, ??ass-ta., ??nya?,??ca!}
you-nom brother-be-adn as patient_V-{imp, pst-dec, int, hor}
'Since you are his brother, {be, you were, were you, let's be} patient!/./?'

Fourth, the main predicate cannot be a stage-level stative:

(59) a. ??Yuna-nun haksayng-i-n mankhum hayngpokha-ta.
-top student-be-adn as happy_V-dec
'Because Yuna is a student, she is happy.'
b. Mino-nun kwunin-i-n mankhum yongkamha-ta.[16]
-top soldier-be-adn as brave_V-dec
'Mino is brave because he is a soldier.'

Hayngpokha 'be happy' is a stage-level predicate, and *yongkamha* 'be brave' is an individual-level predicate. This difference leads to the different acceptability. There are other properties of this construction, but I will not go into them. These properties set the construction from the others involving degrees, and it should be treated separately.

5. Conclusion

In this paper, I considered three types of sentences with *mankhum*. In two of them, a *mankhum*-phrase denotes a degree, regardless of whether the clause in it contains a gradable verb or not. In both cases, a *mankhum*-phrase is an adjunct and it is interpreted with respect to the event denoted by the VP headed by a gradable verb. When a

However, this interrogative is more like a tag question in that the speaker asks for the hearer's confirmation. If the interrogative ends with the normal interrogative ending *-nya*, the sentence becomes odd.

[16] One reviewer came up with this example. I do not know yet why this restriction applies in this construction.

mankhum-phrase includes an adnominal clause that has a gradable verb, a DegP moves to the Spec of a CP. When a *mankhum*-phrase does not include a gradable verb, the phrase is interpreted as a coerced relative clause. The adnominal clause in the *mankhum*-phrase provides an event and the measure function is provided by the matrix clause.

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